

Roadmap

Revision 4 — August 29, 2026. All five adjudications ruled.

Status: **working document**. Changes no rule, records no finding.

The one-line version.

Make the audited state recoverable and the engine startable, build the measurement apparatus the findings themselves presuppose, strip dead complexity, fix the three Criticals, then reshape contracts and output truthfulness — and only then the backtest on-ramp and an independent re-audit. Severity decides which items belong to the gate sets. It does not decide their position.

What changed in Revision 4 — the adjudications are closed.

Viktor ruled on all five open questions on 29 August. The decisions table later in this document now records outcomes rather than open disputes. Three consequences matter more than the rest:

There are now four Critical findings, not three. Item 6 was raised from Major on the principle that severity reflects consequence, not implementation effort. The release gate blocks on four.

The execution order changed. Ruling *degrade* rather than *halt* means sequence item 9 must add fields to the decision object, so the contract that guards field changes has to precede it. Sequence item 10 moves to immediately after sequence item 7. New order: **7 → 10 → 8 → 9 → 11 → 12 → 13 → 14 → 15 → 16**. The item numbers are unchanged — they are citations in four documents.

Position sizing is removed from the engine rather than kept or permitted by amendment. Monetary sizing belongs to a portfolio layer; the engine supplies market-derived risk parameters only.

Two numbering systems, and they collide.

Item N is a rule in the 44-rule register — Item 6 is Traceability, Item 13 is Fail Safely. **Sequence item N** is a step in the sixteen-item roadmap below. They are not the same, and both appear throughout. Where this document means the roadmap it says *sequence item*.

What changed since Revision 1.

Phase A is **done** — the harness exists, runs, and is published. Running it found three defects four audit passes had missed. Step 5 has since run on GLM 5.3 and returned a sixteen-item sequence, which replaces this document's earlier four-phase sketch. Machine learning has been put **on ice** by decision, with written conditions for revisiting. The earlier edition's dateline was wrong and is corrected here.

What changed in Revision 3, and why it exists.

Sequence items 1 and 2 are complete — the audited state is tagged, and the repository now starts from a fresh clone. The test suite moved from 5 passing to 8.

A correction. Revision 2's open-decisions table listed five rulings and one of them was not the reviewer's. Claude's summary of Step 5 dropped the position-sizing question — which Step 5 names explicitly in its own opening paragraph — and substituted the halt-or-degrade question, which Step 5 raises elsewhere. Both are real; there are five, and the missing one is now restored below. Anyone reading Revision 2 to make these rulings would have made four and never learned of the fifth. Recorded here rather than quietly corrected, per this project's practice.

Where the project stands today

The Constitution was ratified on August 26 and audited on August 27 — four runs through OpenRouter, using models with no prior involvement in the project, at roughly one dollar each. All 44 rules carry a recorded finding. The scope freeze lifted when the last Tier 1 item was graded, and three amendments were adopted the same day.

	Compliant	Non-compliant	Unknown	Total
Tier 1 — invariants	10	10	1	21
Tier 2 — architecture	3	4	0	7
Tier 3 — process	3	3	4	10
Tier 4 — preferences	5	0	1	6
Total	21	17	6	44

The four Critical findings

Three came from the audit. Item 6 was raised from Major to Critical by Viktor's ruling of 29 August — on the principle that severity reflects consequence, not implementation effort. The fix is one line; the consequence is that the panel asserts a safety action which did not occur, on every run.

Item 3 — Data Integrity

Nothing detects missing candles, duplicates, impossible prices, bad timestamp ordering, stale data or abnormal volume. Defects are silently filled in by ffill/bfill rather than caught, so the engine cannot tell “no defect found” from “defect fabricated away.”

Item 6 — Traceability **[RAISED TO CRITICAL 29 AUG]**

The panel prints “Trade logged to Logs/phase7_trade_log_<symbol>.csv” on every run and no code anywhere writes that file. The engine reports an audit action as having happened when it did not — and “why did it decide that?” is unanswerable once the process exits.

Item 11 — No Circular Reasoning

trend_health is counted at least four times: 30% of bias_score, again directly in confidence, again as the whole base of validation_score, and again as Current Market. The panel presents one number as four agreeing signals.

Item 13 — Fail Safely

When an indicator fails, the engine substitutes confident-looking constants — RSI 50, ADX 25, SuperTrend direction 1.0 — with no marker anywhere. A failed SuperTrend silently adds a permanent +15 bullish vote to the bias score.

The release gate now in force — and it blocks on four.

No output of this engine may be relied on for a real trading decision while any Critical Tier 1 finding stands unresolved, and backtesting does not begin until Items 2, 3, 6 and 18 are all Compliant. Running the engine to look at is fine. Acting on it is not.

Item 6's promotion moves it from optional to mandatory in the minimum gate-opening set. It was already required for the *backtest* gate; it now blocks the release gate too.

What Phase A found, and why it changed the ordering

The harness was built to make the fixes safe. Its first real run found defects instead — three that four audit passes across three models had all missed, because every one of those passes read the code and none of them ran it.

Finding	Status	Where it lands
The repository does not start from a fresh clone. main.py builds its log handler at module scope (line 16) and creates Logs/ inside main() (line 41). FileHandler opens eagerly, so import fails before the try/except that would catch it.	Severity correction — Run 1 found the ordering and called it soft. It is not soft.	Item 2 of the sequence, merged into T2-6
AERO is hardcoded into user-facing reasoning text at decision_model.py 411, 413 and 419, and panel_render.py:83. Run the engine on SOLUSDT and the explanation talks about AERO; run it on BTCUSDT and it compares BTC to itself.	Unrated — no register run examined explanation-text provenance. Contradicts Run C's Compliant on T4-2: true of the arithmetic, false of the prose.	Item 12, under Item 6 (traceability)
"relationship relationship" — decision_model.py:419 appends the word to a label that already ends in it.	Unrated. Minor, cosmetic, one line.	Item 12, rides the same strings

The lesson is not that three models read badly. It is that reading and running are different instruments, and the project had been buying more of the first while owning none of the second.

The sequence

Sixteen items, produced by Step 5 and verified against source. The organising principle is not severity: **position is set by what has to exist before a fix can be proven correct, and by which edits share a seam**. Severity decides membership in the gate sets.

The sentence the whole ordering rests on.

A fix made before its apparatus exists can be *made*, never *resolved* — because “resolved” under the finding schema requires a Verification field, and the release gate requires resolution. That is why the apparatus phase is not process hygiene sitting in front of the real work. It is the gate-opener.

0 Adjudications and registrations

PROCESS — VIKTOR

Five rulings, two of which change the ordering — see the full table later in this document. (Revision 2 said four here and listed the wrong set; corrected in Revision 3.) The two unrated harness findings enter the formal record here or get lost the way soft-rated defects already were.

Depends on nothing. **Unblocks** stable positions for items 12 and 13, and the minimum gate-opening set. **Effort** small.

1 T3-7 — a known-good checkpoint

DONE — 29 AUG

One git tag on the audited commit. The repository had 23 commits and zero tags, so there was no return point at all, and re-audit diffs had nothing to compare against.

Done. Tag audited-2026-08-27 points at 19516e1 — the last commit before any engine source change. The tag is exact rather than approximate: nothing committed between the audit closing and that point touched engine source.

2 T2-6 — controlled dependencies, and the clone failure

DONE — 29 AUG

Two mechanisms, one outcome. requirements.txt omitted requests (which data_fetcher imports, so the first import failed) and colorama; it declared ccxt, which appeared nowhere else in the codebase and is execution-capable. Separately, the log handler was built before the directory existed.

Done in cc7f8ff. The makedirs sits above the model imports rather than immediately above basicConfig, because live_trading instantiates its simulator at module scope and that constructor also touches the filesystem. The failure and the fix were reproduced in isolation before the edit was written.

Three tests flipped green — both clean-checkout tests and the module-import check. A fourth, the dependency check, broke and was repaired in the same commit: the previous commit had added reportlab build scripts to a repository whose test walker scans every Python file, so the harness reported reportlab as an undeclared engine dependency. *The harness caught a defect introduced one commit earlier — which is the argument for building it first, demonstrated rather than asserted.*

3 Pinned-data path and archived datasets

APPARATUS

load_csv exists but nothing wires it in — get_tf always fetches live, so the engine cannot currently produce two comparable runs, and even T2-4's own prescribed verification is impossible.

Inject a source at get_tf covering all three series the engine fetches — base, macro 1d, and BTC context — or runs stay network-dependent. **Effort** small-medium. Closes T3-5's Unknown as a side effect.

4 T3-3 / T3-4 stage 1 — bring the harness in

APPARATUS

The harness currently lives outside the nineteen-file source. Bring it in as the suite skeleton, add a smoke run on pinned data and grep guards for ccxt and credential patterns, and begin commit-per-change with dated notes — the remediation then generates the evidence that later closes T3-1, T3-2 and T3-8.

No CI exists in the repository. Do not assume it; a locally green suite is the standard. **Effort** medium.

5 Item 16 — delete unconsumed complexity

APPARATUS

Before the golden pin, so the baseline does not enshrine dead code. Delete the Bollinger trio, KAMA and its slope, DIP/DIM, Typical_Price, the unreachable VWAP plot branch, compute_exit, the dead TARGET-HIT colour branches, the unreachable hysteresis branches in _detect_regime, and the stale roadmap comments describing code that no longer exists.

VWMA stays — entry_model consumes it for distance scoring. **Verification:** deleting unconsumed code is provable by output-invariance on pinned data. **Effort** small-medium.

6 T2-1 and cache removal — shared-frame aliasing

APPARATUS

calculate_dynamic_bias rewrites its caller's columns in place. The indicator cache mutates cached frames, and its key embeds the last close in a per-process dict — so it can never hit across runs at all.

Deleting the cache is cheaper than repairing the key, and it dissolves the open Items 4 / 12 dispute by repair rather than adjudication. **Effort** small.

7 T3-4 stage 2 — the golden baseline

APPARATUS

Captured after cleanup and before any semantic change, so every later delta is attributable to exactly one controlled change. Exclude chart PNGs (unstable hashes); make any uninjected fetch fail deterministically — the data_fetcher singleton binds base_url at import, so patch the instance, not just config.

Effort small. Without this, the Item 11 and 13 fixes are unprovable as “changed exactly as intended.”

8 Item 3 — Data Integrity

CRITICAL

First Critical because it is upstream: no downstream number is trustworthy while garbage enters silently. The validation gate must sit in `fetch_ohlcv` **before** the `close_time` column is discarded — the raw response carries it and throws it away one line later, and staleness checks need it.

Reject, do not fabricate. A failed macro timeframe renders UNAVAILABLE, never NEUTRAL — a fabricated neutral is a directional vote at 10% weight. **Effort** medium. The eight corrupted fixtures are already written.

9 Items 13 + 8 — fail safely, merged

CRITICAL

One defect family reported under two invariants. Delete the fabrication constants — RSI 50, ADX 25, ATR at 2% of price, SuperTrend = close, ST_Direction 1.0, the trend_health 50.0 defaults, risk_model's direction-blind except-return — and let failures reach the existing error path.

A decision for Viktor sits here: halt on failure, or degrade with a flag and cut confidence. Step 5 declined to choose, on the stated grounds that its preference for the simpler branch is a habit it shares with the model family that built this engine. **Effort** medium.

10 T2-3 — the decision-object contract

SHAPE

Between the shape-neutral Criticals and the reshaping items, because 11 through 13 add, remove and rename fields — and renaming without a net is this codebase's demonstrated failure mode: a rename once broke fourteen modules at once.

mypy does not exist in the repository. Adding it is apparatus belonging to this item. **Effort** medium.

11 Item 11 — no circular reasoning

CRITICAL

Third Critical, sequenced after 8 and 9 so the score being de-circularised is computed on validated, non-fabricated data — otherwise the golden deltas conflate two causes. Remove `trend_health` from `_compute_confidence` (it is already inside `bias_strength` at 0.30), rebase `validation_score`, and delete the redundant renderings rather than inventing distinct metrics.

Coupling rule: the reason strings change in the same commit. Prose describing the old formula is an Item 8 regression the moment the number changes. **Effort** medium.

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Items 5 + 6 — reproducibility and traceability

SHAPE

Implement the decision log the panel already claims exists. Make “Trade logged” name a file that provably exists; fix the chart path printing None; stamp engine_version, which exists in config and is written nowhere; add Item 7's epistemic label; replace the hardcoded AERO with the run's symbol; fix the doubled word.

Position is conditional on the Item 6 ruling — sequenced here under the stricter reading. A Major ruling swaps this with item 13. **Effort** medium.

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Item 14 — risk is not conviction

SHAPE

Remove the risk_score / signal_strength aliases; remove bias_factor from stop_mult — conviction currently tightens stops arithmetically, which is Item 14's prohibition written as code; wire volatility_state into calculate_stop_targets, which engine_core omits, leaving the HIGH and EXTREME widening tiers permanently inert; normalise the inverted entry zone at source.

Effort small-medium. Stop distances will change — intended, but review it as a scan diff rather than merging silently.

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T2-4 — explicit configuration

SHAPE

After the deletions, so knobs are not wired to indicators just removed. Config declares CHART_HEIGHT 10 and CHART_DPI 150 while plotting hardcodes 8 and 200 — wiring it changes chart artifacts, visibly and intentionally.

Effort small; near-zero risk on the decision path, since config values equal the current hardcodes. Assert via golden invariance.

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Item 2's standing caveat — bfill quarantine

BACKTEST ON-RAMP

Last code item, because nothing in the live path reads it — the decision row is always the last bar — and because its verification apparatus, re-computing historical decisions on truncated data, exists in no form and is itself half a backtester.

The backtest gate's letter does not require Items 11 and 13 fixed. This plan fixes them earlier anyway: backtesting an engine with double-counted confidence manufactures precisely the false validation Item 7 exists to prevent. **Effort** medium.

1 Step 8 — independent re-audit, scoped to what 6 changed

RE-AUDIT

Terminal, and it belongs to an independent auditor rather than the party that made the fixes. Scope it to changed items with artifact evidence — panel screenshots, test output, log files — per the Evidence discipline.

This is where the release gate formally opens. Item 15 stays Unknown until backtesting exists; that is the honest status, not a lapse. **Effort** small.

Four findings that are one defect reported twice

The invariants stay distinct for re-audit purposes. The *edits* merge — which is why seventeen Non-compliances do not mean seventeen pieces of work.

Reported under	The single underlying defect	Fixed at
Items 5, 6 and 8(b)	The phantom trade log — the panel announces a CSV that no code writes, every run.	item 12
Item 3, Item 8(a), Item 13, blind-run finding 2	The fabricated fallback constants.	item 9
Item 10(c) and Item 14(a)	The risk_score / signal_strength aliasing.	item 13
Item 11, Item 10(a), Item 14(b)	trend_health overloading — one quantity doing four jobs, rendered three times, and the validation gate rebasing that follows from fixing it.	item 11

Item 8 — Epistemic Honesty — has no unique remediation work of its own. It is the parent principle whose every concrete instance is co-reported elsewhere, and its re-audit rides items 9 and 12.

Five things cheaper to delete than to repair

Deletion is a legitimate remedy, not a shortcut — and all five are git-reversible and provable by output-invariance.

Item 16's dead indicators — wiring them would be new-feature work.

compute_exit — wiring it presupposes a held-position concept the engine deliberately lacks, and exit_watch already performs the advisory role.

The indicator and structure caches — they never hit across runs; deletion removes the stale-serving hazard and the mutation hazard at zero functional cost.

The fabricated fallbacks — cheaper than building a degraded-mode subsystem — though this is the decision Step 5 handed to Viktor rather than making.

The redundant MOMENTUM and Current-Market lines — deleting is honest; inventing distinct underlying metrics to justify them is new-feature work.

The adjudications — RULED, 29 August

All five closed by Viktor on 29 August. Under Roles & Authority these were his alone; Claude's job was to lay out the disagreement and the consequences, not to choose. Recorded here as outcomes rather than open questions.

Question	Ruling	What it changed
Item 6 severity	CRITICAL. “The remediation is small, but severity reflects consequence, not implementation effort.”	Four Criticals now, not three. Item 6 becomes mandatory in the minimum gate-opening set. No reordering — the sequence was already built under the stricter reading.
Position sizing <i>constitutional, not code</i>	REMOVE FROM THE ENGINE. “Monetary position sizing belongs outside the Structural Quant Engine, in the portfolio/execution layer. The engine may provide market-derived risk parameters such as entry, stop, and volatility, but it should not decide how many dollars to allocate.”	Five fields leave the decision object and calculate_position_size is deleted. No Constitution amendment needed — removal aligns the engine with Item 14's existing rationale rather than changing it. Lands at sequence item 13.
Item 13 — halt or degrade	DEGRADE. “It must not fabricate replacement values. The failure must be recorded in the decision output, and confidence and trade quality must be reduced accordingly. A degraded result does not by itself authorize trading.”	Changed the execution order — see the box below. Also enlarges sequence item 9: it now needs a degraded-state field, a record of which indicators failed, and a gate preventing a degraded run from authorising a trade.
Item 2 strength	COMPLIANT, rationale amended. “Input-side backward fill can propagate through recursive indicators and may affect the final decision.”	Sequence item 15 goes from advisory to mandatory for the backtest gate. No reordering. Amends a finding's reasoning, not a rule — no amendment-control process.
Items 4 and 12	DISSOLVED BY REMEDIATION. Resolved by deleting the caches at sequence item 6.	No separate repair, no adjudication spent. Repair rather than argument makes both readings true.
Item 20 amendment	STILL OPEN. The only remaining one.	Blocked on a reviewer who is not Claude. Complicated by Gemini and Copilot both refusing to ingest the Constitution PDF — a content-classification false positive.

The ordering consequence of ruling “degrade”.

Step 5 placed sequence item 10 — the decision-object contract — *after* the Criticals, and gave a specific reason: “Between the shape-neutral Criticals (8/9 use the existing error path; no new fields) and the reshaping clusters (11–13 add, remove, and rename fields).”

That premise held only under “halt”. Degrading requires new fields, so sequence item 9 is no longer shape-neutral, and under the original order it would reshape the decision object before the contract guarding field changes exists — in a codebase whose own record shows a rename once broke fourteen modules at once.

Sequence item 10 therefore moves to immediately after sequence item 7. It depends only on item 7, so nothing is violated, and placing it ahead of item 8 as well is free — the data-integrity validator will likely need to report *why* it rejected input, which is also a field.

New execution order: 7 → 10 → 8 → 9 → 11 → 12 → 13 → 14 → 15 → 16. The item numbers themselves do not change; they are citations across four documents.

Splitting item 9 was considered and rejected. The split would delete the fabrication constants first — shape-neutral — and add degraded-state plumbing later. But deleting the constants requires something to catch the resulting failures, and before the degraded path exists the only mechanism available is the old error path, which halts. The split would ship, however briefly, exactly the behaviour that was ruled against.

A trap in the position-sizing removal — verified in source.

`risk_amount` names two unrelated things. In `engine_core.py:827` it is money — `account_balance × risk_percent` — and that is what gets deleted. In `panel_render.py:1078` it is a price distance, `abs(current_price - stop_loss)`, and it is the denominator for all three R:R ratios on the panel.

A find-and-replace on the name will turn “TARGET 1 | R:R 1 : 1.00” into zero, silently. The removal is five fields from `signal_router`, three computations in `engine_core`, and `calculate_position_size` from `risk_model` — nothing else consumes them.

The smallest set that opens the release gate.

The clone fix (sequence item 2 — **done**); the thin apparatus slice — pinned dataset, corrupted fixtures, fault injection, one golden baseline; the decision-object contract, now moved ahead of the Criticals; then Items 3, 13, 11 **and** 6. Then a scoped independent re-audit, because “resolved” is not self-declared.

Item 6 is no longer optional in this set. Revision 3 listed it as conditional on the severity ruling, with a Major finding removing it at zero cost. The ruling came back Critical, so it is required. The honesty caveat that used to sit here — include the one-line fix regardless, because the panel otherwise lies on the first run anyone relies on — is now the rule rather than an exception to it.

Parked by decision on August 29, 2026 — not cancelled.

The conditions for revisiting are written down below so that the decision can be revisited on evidence rather than on enthusiasm. This section is kept in full because the reasoning is the useful part.

Can you add an LLM?

Yes, easily. It would read the panel and explain it. But the panel currently shows one number three times and calls it three signals — so the LLM would confidently describe agreement that isn't there. It would hide the bugs, not find them.

Architecturally it is straightforward: an LLM layer sits outside the decision path and touches no invariant about execution or authority. The problem is epistemic rather than technical. Fluent commentary over defective numbers is worse than no commentary, because it makes the defect harder to notice. This becomes reasonable after item 11, not before it.

Can you add machine learning?

Yes, eventually. The obvious use is fitting your six bias weights (0.30 / 0.20 / 0.15 / 0.15 / 0.10 / 0.10) against real outcomes instead of choosing them by hand.

Those weights are currently judgment calls. Fitting them is a legitimate, modest, testable application — and it needs labelled outcomes, which need a backtester. There is no shortcut around that ordering.

The trap that kills these projects.

Training a model means evaluating at historical decision points, which is precisely what converts the dormant backward-fill leak into a live one. The model would learn from information that did not exist at the time. It would look excellent in testing and be worthless in use — and the failure is silent, because a leaking model produces beautiful results rather than obvious errors. The leak is already installed and already documented. Removing it is cheap now and expensive to diagnose later.

Three further reasons, found by checking the source

The training data does not exist. `live_trading.py:210` does persist a decision as a JSON file, but nothing anywhere records what price did afterward, and `engine_core.py:466` overwrites its state file each run rather than accumulating. Decisions without outcomes are unlabelled examples.

Item 11 makes fitted weights worse, not better. Fitting over features that are secretly the same feature produces large offsetting coefficients that swing on tiny data changes. Hand-picked weights on collinear inputs are crude; fitted weights on collinear inputs are crude and wearing a lab coat.

Item 13's sentinels would become learned signal. `RSI = 50` means the indicator broke. A model trained on that learns to predict from failure markers.

Conditions for taking it off ice — all five

1. Items 2, 3, 11 and 13 fixed **and verified by the harness**.
2. A working decision *and* outcome log, with real accumulated history.
3. A backtester validated against known-answer cases.
4. A kill condition written down **before** the first fit — already a Constitution amendment candidate.
5. An out-of-sample test that beats the current hand weights. If it does not, that is a real result: it means the hand weights were fine.

What to stay suspicious of.

Machine learning feels like the milestone that makes a project serious. That instinct is precisely what the Constitution was written to catch. It is also worth noting that fitting six coefficients is a small regression, grid-searchable by hand once outcome data exists — calling it machine learning oversells what it is. ML earns its place only for interactions the hand weights cannot express, and that needs far more data than six weights do.

What has to happen to the Constitution before any of this

Nineteen proposed invariants were declined on August 27 — multiple-testing correction, out-of-sample validation, calibration monitoring, regime and distribution-shift handling, statistical uncertainty on point estimates — on the correct grounds that they constrain a system with backtesting and execution that this engine does not have.

Every one of them becomes load-bearing the day machine learning enters. Feature selection and parameter search across a few hundred four-hour candles is a machine for manufacturing false discoveries, and those rules are the defence against it. They are recorded in Future Amendment Candidates. Pull them back in *before* the first line of model code, not after the first promising result — the point at which a fitted model looks good is the worst possible moment to start negotiating with the rules that would invalidate it.

What this roadmap deliberately does not include

No new features. Nothing is added to the engine anywhere in items 1 through 15 — every one is a repair, a test, or a piece of measurement apparatus. The engine gets smaller and more honest before it gets larger.

No date estimates. The engine was built fast and audited once. Guessing how long sixteen items take, in a codebase whose test suite is four days old, would be the same kind of confident number the audit spent all day objecting to. Effort is stated as small, medium or large and nothing finer.

No claim that this ordering is the only defensible one. Step 5 named two places where a second ordering is arguable and gave the trade-off rather than picking silently: doing Item 11 before the decision-object contract would open the release gate one item sooner, at the cost of renaming fields without a net in the codebase where exactly that once broke fourteen modules.

The sequence is the point. Each item exists to make the next one measurable — the apparatus makes the fixes provable, the fixes make the backtester meaningful, and the backtester is what would make machine learning something other than a guess with better presentation.